

Name: _____

Class: _____

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Binary Shifts

Work in pencil. Every bit moves the same number of columns, all together. The empty columns left behind are filled with **0**. Anything pushed off either end is **lost**.

The two rules

Left shift — every bit moves *left*.

Each column left is worth *twice* as much, so a shift of n places **multiplies by 2^n** .

1 place $\times 2$ 2 places $\times 4$ 3 places $\times 8$

Right shift — every bit moves *right*.

Each column right is worth *half* as much, so a shift of n places **divides by 2^n** .

1 place $\div 2$ 2 places $\div 4$ 3 places $\div 8$

Worked example 1 22, shifted LEFT 1 place

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	0	0	1	0	1	1	0	22
Answer	0	0	1	0	1	1	0	0	44
Check			32		8	4			

The **Check** row is the proof: the 1s now sit in the 32, 8 and 4 columns, and $32 + 8 + 4 = 44$. The original was 22, so the shift has multiplied by 2.

Worked example 2 20, shifted RIGHT 1 place

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	0	0	1	0	1	0	0	20
Answer	0	0	0	0	1	0	1	0	10
Check					8		2		

$8 + 2 = 10$, and the original was 20 — divided by 2.

What “lost” means, and it is different at each end.

A 1 pushed off the **left** end is gone because the answer is now too big for eight bits — that is an **overflow**, and the number stored is wrong.

A 1 pushed off the **right** end is gone because the answer is not a whole number — the fraction is thrown away. 25 shifted right 1 place gives **12**, not 12.5.

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Left shifts1. Shift **13 LEFT** by **1** place.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	0	0	0	1	1	0	1	13
Answer	0	0	0	1	1	0	1	0	26
Check				16	8		2		

2. Shift **37 LEFT** by **2** places.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	0	1	0	0	1	0	1	37
Answer	1	0	0	1	0	1	0	0	148
Check	128			16		4			

3. Shift **100 LEFT** by **2** places.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	1	1	0	0	1	0	0	100
Answer	1	0	0	1	0	0	0	0	144
Check	128			16					

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Right shifts4. Shift **44 RIGHT** by 1 place.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	0	1	0	1	1	0	0	44
Answer	0	0	0	1	0	1	1	0	22
Check				16		4	2		

5. Shift **200 RIGHT** by 2 places.

Place value	128	64	32	16	8	4	2	1	Denary
Number	1	1	0	0	1	0	0	0	200
Answer	0	0	1	1	0	0	1	0	50
Check			32	16			2		

6. Shift **25 RIGHT** by 1 place.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	0	0	1	1	0	0	1	25
Answer	0	0	0	0	1	1	0	0	12
Check					8	4			

What went wrong

Questions 3 and 6 did not give the answer the arithmetic says they should. Both lost a bit — but off *opposite ends* of the grid, and that is the whole difference.

7. Look back at **question 3**. $100 \times 4 = 400$, but the grid does not say 400. What has happened, and what is it called? **Overflow.** 400 needs nine bits. The 1 that was in the 64s column should have moved to a 256s column, and there is not one — it was pushed off the LEFT end and lost. The grid reads 144, which is $400 - 256$.

8. Look back at **question 6**. $25 \div 2 = 12.5$, but the grid does not say 12.5. What has happened here, and is it the same problem as question 3? **Loss of precision — and no, it is the opposite end.** The 1 in the 1s column was pushed off the RIGHT and lost. Binary integers cannot hold a half, so the fraction is thrown away and the answer is rounded down to 12. Question 3 lost a bit off the left, which made the number too big to store; this loses a bit off the right, which makes it not exact.

9. A programmer needs to multiply by 8 very quickly. Which shift should they use, and in which direction? **A left shift of 3 places, because $2^3 = 8$.**